5. Find the elasticity of demand in terms of x for the following demand laws and also find the output (x), when the elasticity is equal to unity

   (i) $p=(a-bx)^2$ (ii) $p=a-bx^2$

6. Find the elasticity of supply for the supply function $x=2p^2+5$ when p=3

7. The demand curve of a commodity is given by $p=\frac{50-x}{5}$, find the marginal revenue for any output x and also find marginal revenue at $x=0$ and $x=25$?

8. The supply function of certain goods is given by $x=a\sqrt{p-b}$ where p is unit price, a and b are constants with $p>b$. Find the elasticity of supply at $p=2b$.

9. Show that $MR=p\left[1-\frac{1}{\eta_d}\right]$ for the demand function $p=400-2x-3x^2$ where p is unit price and x is quantity demand.

10. For the demand function $p=550-3x-6x^2$ where x is quantity demand and p is unit price. Show that $MR=p\left[1-\frac{1}{\eta_d}\right]$

11. For the demand function $x=\frac{25}{p^4}$, $1\le p\le 5$, determine the elasticity of demand.

12. The demand function of a commodity is $p=200-\frac{x}{100}$ and its cost is $C=40x+120$ where p is a unit price in rupees and x is the number of units produced and sold. Determine (i) profit function (ii) average profit at an output of 10 units (iii) marginal profit at an output of 10 units and (iv) marginal average profit at an output of 10 units.

13. Find the values of x, when the marginal function of $y=x^3+10x^2-48x+8$ is twice the x.

14. The total cost function y for x units is given by $y=3x\left(\frac{x+7}{x+5}\right)+5$. Show that the marginal cost [MC] decreases continuously as the output (x) increases.

15. Find the price elasticity of demand for the demand function $x=10-p$ where x is the demand and p is the price. Examine whether the demand is elastic, inelastic or unit elastic at $p=6$.

16. Find the equilibrium price and equilibrium quantity for the following functions.

    Demand: $x=100-2p$ and supply: $x=3p-50$

17. The demand and cost functions of a firm are $x=6000-30p$ and $C=72000+60x$ respectively. Find the level of output and price at which the profit is maximum.

18. The cost function of a firm is $C=x^3-12x^2+48x$. Find the level of output $(x>0)$ at which average cost is minimum.

6.2 Maxima and Minima

We are using maxima and minima in our daily life as well as in every field such as chemistry, physics, engineering and in economics etc.,

In particular, we can use maxima and minima

(i) To maximize the beneficial values like profit, efficiency, output of a company etc.,

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